untitled

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```
from itertools import permutations

def solve(words, result):
    letters = sorted(set("".join(words) + result))
    if len(letters) > 10:
        raise ValueError("More than 10 distinct letters: no valid assignment.")
    leading = {w[0] for w in words + [result]}

    def value(word, mapping):
        n = 0
        for ch in word:
            n = n * 10 + mapping[ch]
        return n

    solutions = []
    for digits in permutations(range(10), len(letters)):
        mapping = dict(zip(letters, digits))
        if any(mapping[ch] == 0 for ch in leading):
            continue
        if sum(value(w, mapping) for w in words) == value(result, mapping):
            solutions.append(mapping)
    return solutions

def show(words, result, mapping):
    def num(w):
        return "".join(str(mapping[c]) for c in w)
    width = len(num(result)) + 2
    for i, w in enumerate(words):
        sign = "+" if i == len(words) - 1 else " "
        print(f"{sign} {num(w):>{width}}")
    print("-" * (width + 2))
    print(f"  {num(result):>{width}}")

if __name__ == "__main__":
    puzzles = [(["SEND", "MORE"], "MONEY"),
               (["TO", "GO"], "OUT"),
               (["BASE", "BALL"], "GAMES")]
    for words, result in puzzles:
        print(" + ".join(words), "=", result)
        sols = solve(words, result)
        if not sols:
            print("No solution.\n")
            continue
        print("Letter mapping:", dict(sorted(sols[0].items())))
        show(words, result, sols[0])
        print(f"({len(sols)} solution(s) found)\n")
```

Ln: 46 Col: 51

26°C Light rain

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```
SEND + MORE = MONEY
Letter mapping: {'D': 7, 'E': 5, 'M': 1, 'N': 6, 'O': 0, 'R': 8, 'S': 9, 'Y': 2}
     9567
+    1085
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    10652
(1 solution(s) found)

TO + GO = OUT
Letter mapping: {'G': 8, 'O': 1, 'T': 2, 'U': 0}
     21
+    81
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    102
(1 solution(s) found)

BASE + BALL = GAMES
Letter mapping: {'A': 4, 'B': 7, 'E': 3, 'G': 1, 'L': 5, 'M': 9, 'S': 8}
     7483
+    7455
---------
    14938
(1 solution(s) found)
```